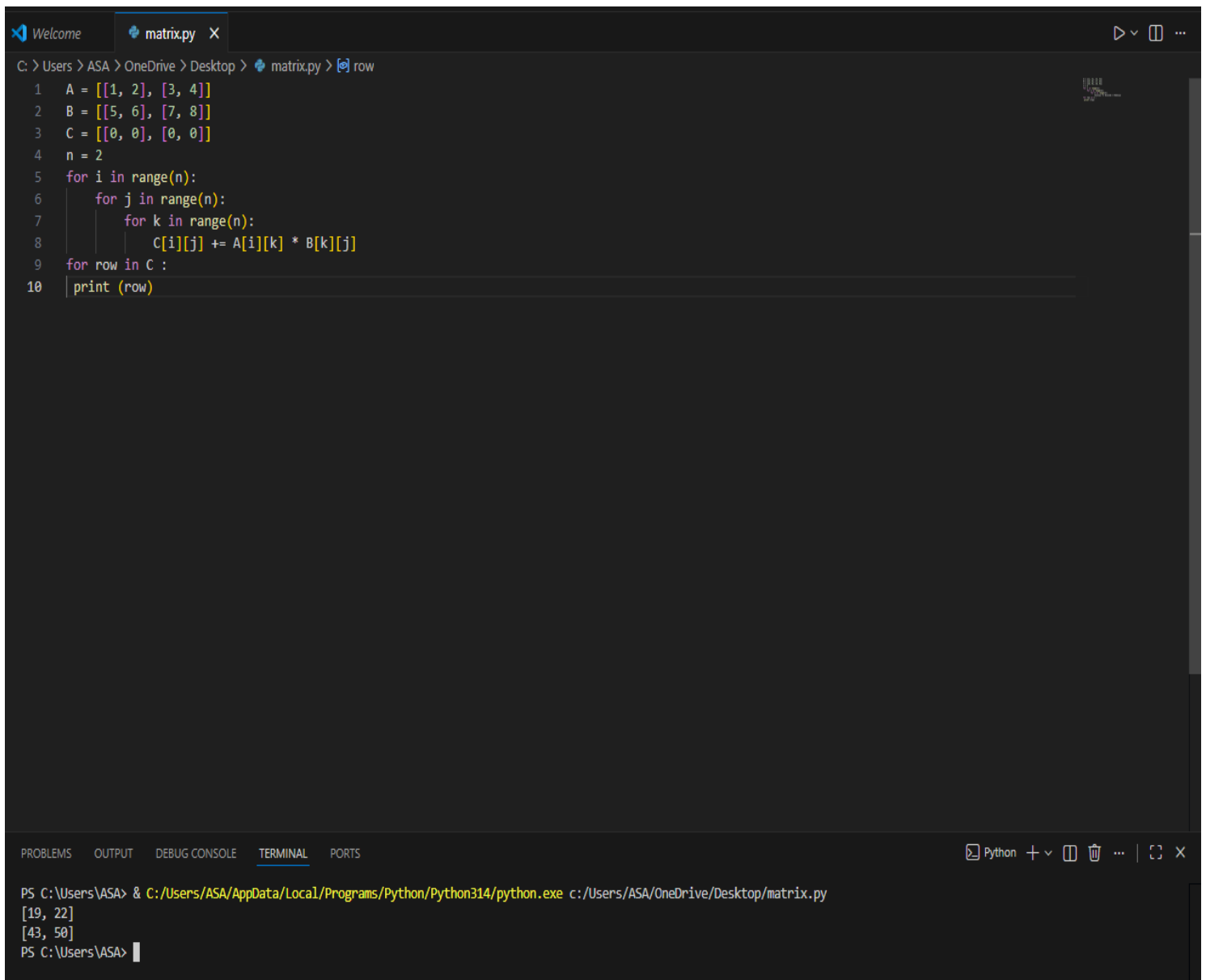


نام : شایان عباسی

تکالیف شماره 2

درس برنامه نویسی پیشرفته

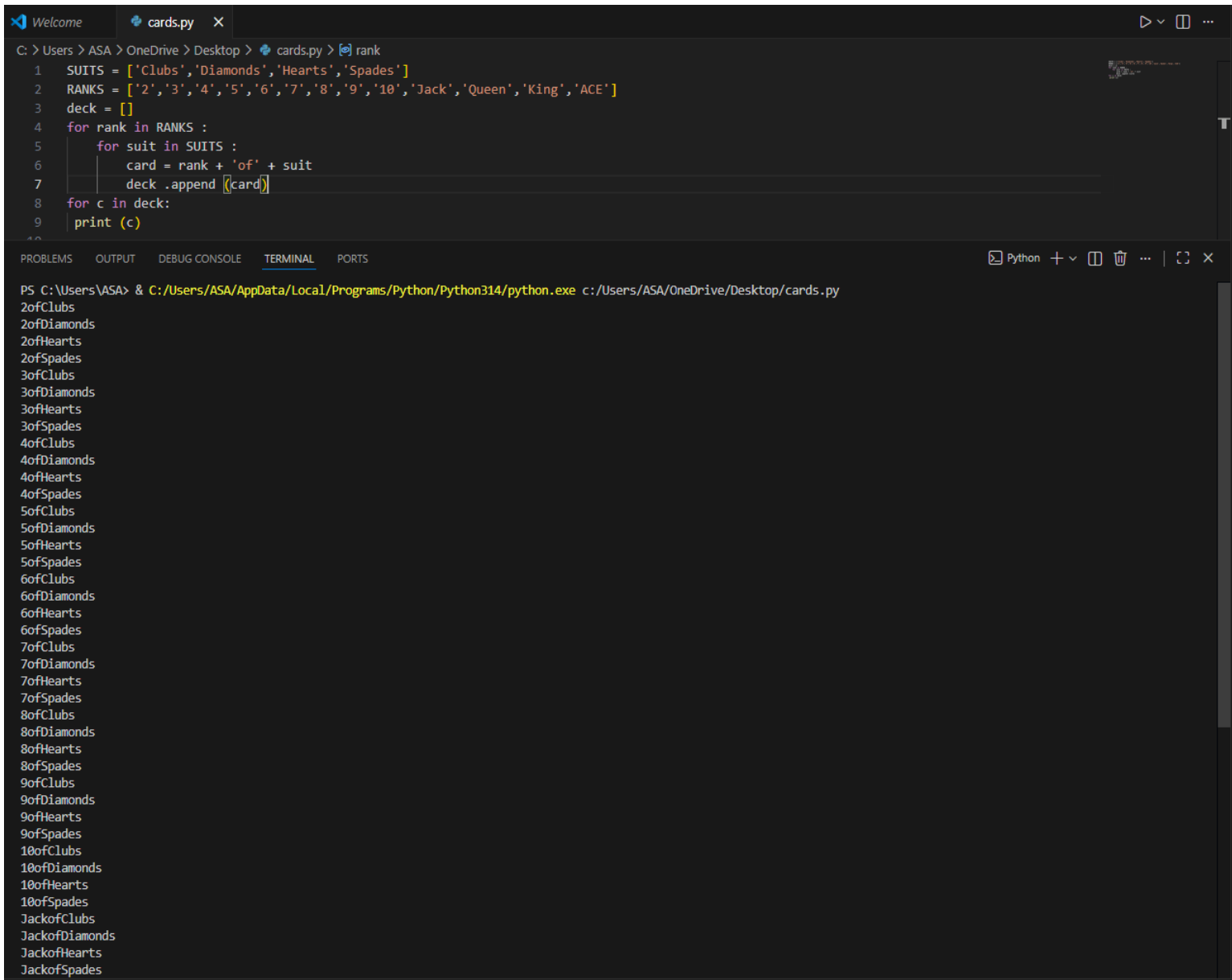
مثال ضرب ماتریسی (2×2)



```
matrix.py X
C: > Users > ASA > OneDrive > Desktop > matrix.py > row
1 A = [[1, 2], [3, 4]]
2 B = [[5, 6], [7, 8]]
3 C = [[0, 0], [0, 0]]
4 n = 2
5 for i in range(n):
6     for j in range(n):
7         for k in range(n):
8             C[i][j] += A[i][k] * B[k][j]
9 for row in C :
10 | print (row)
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python + v [trash] [refresh] [close]
PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/matrix.py
[19, 22]
[43, 50]
PS C:\Users\ASA> |
```

مثال ترتیب چیدمان کارت ها



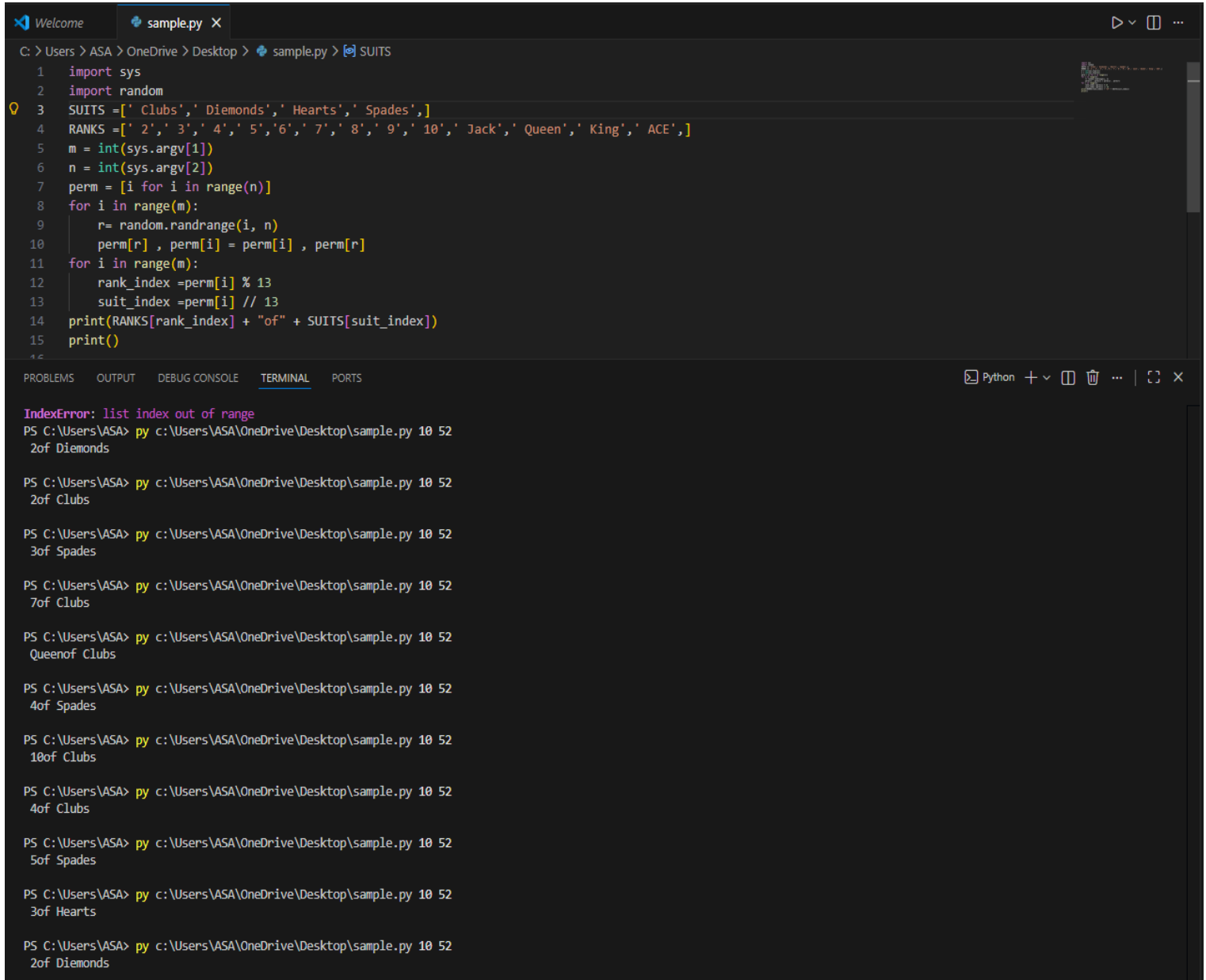
The image shows a Visual Studio Code editor window with a file named `cards.py` open. The script defines a list of suits, a list of ranks, and a deck. It uses nested loops to generate a list of cards in the format "rank of suit". The terminal output shows the cards in order: 2 of Clubs, 2 of Diamonds, 2 of Hearts, 2 of Spades, 3 of Clubs, 3 of Diamonds, 3 of Hearts, 3 of Spades, 4 of Clubs, 4 of Diamonds, 4 of Hearts, 4 of Spades, 5 of Clubs, 5 of Diamonds, 5 of Hearts, 5 of Spades, 6 of Clubs, 6 of Diamonds, 6 of Hearts, 6 of Spades, 7 of Clubs, 7 of Diamonds, 7 of Hearts, 7 of Spades, 8 of Clubs, 8 of Diamonds, 8 of Hearts, 8 of Spades, 9 of Clubs, 9 of Diamonds, 9 of Hearts, 9 of Spades, 10 of Clubs, 10 of Diamonds, 10 of Hearts, 10 of Spades, Jack of Clubs, Jack of Diamonds, Jack of Hearts, and Jack of Spades.

```
1 SUITS = ['Clubs', 'Diamonds', 'Hearts', 'Spades']
2 RANKS = ['2', '3', '4', '5', '6', '7', '8', '9', '10', 'Jack', 'Queen', 'King', 'ACE']
3 deck = []
4 for rank in RANKS :
5     for suit in SUITS :
6         card = rank + ' of ' + suit
7         deck.append(card)
8 for c in deck:
9     print(c)
```

PS C:\Users\ASA> & C:/Users/ASA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASA/OneDrive/Desktop/cards.py

2 of Clubs
2 of Diamonds
2 of Hearts
2 of Spades
3 of Clubs
3 of Diamonds
3 of Hearts
3 of Spades
4 of Clubs
4 of Diamonds
4 of Hearts
4 of Spades
5 of Clubs
5 of Diamonds
5 of Hearts
5 of Spades
6 of Clubs
6 of Diamonds
6 of Hearts
6 of Spades
7 of Clubs
7 of Diamonds
7 of Hearts
7 of Spades
8 of Clubs
8 of Diamonds
8 of Hearts
8 of Spades
9 of Clubs
9 of Diamonds
9 of Hearts
9 of Spades
10 of Clubs
10 of Diamonds
10 of Hearts
10 of Spades
Jack of Clubs
Jack of Diamonds
Jack of Hearts
Jack of Spades

برنامه انتخاب کارت بدون تکرار



The image shows a Visual Studio Code editor window with a Python file named `sample.py` open. The file contains a script that generates a random permutation of 13 cards (2 through 10 of Clubs, Diamonds, Hearts, and Spades) and prints them in order. The script uses `sys.argv` to take command-line arguments for the number of cards (`m`) and the number of suits (`n`).

```
1 import sys
2 import random
3 SUITS = ['Clubs', 'Diamonds', 'Hearts', 'Spades',]
4 RANKS = ['2', '3', '4', '5', '6', '7', '8', '9', '10', 'Jack', 'Queen', 'King', 'ACE',]
5 m = int(sys.argv[1])
6 n = int(sys.argv[2])
7 perm = [i for i in range(n)]
8 for i in range(m):
9     r = random.randrange(i, n)
10    perm[r], perm[i] = perm[i], perm[r]
11 for i in range(m):
12     rank_index = perm[i] % 13
13     suit_index = perm[i] // 13
14 print(RANKS[rank_index] + " of " + SUITS[suit_index])
15 print()
```

The terminal output shows the execution of the script with the following commands and results:

```
PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
2 of Diamonds

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
2 of Clubs

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
3 of Spades

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
7 of Clubs

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
Queen of Clubs

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
4 of Spades

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
10 of Clubs

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
4 of Clubs

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
5 of Spades

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
3 of Hearts

PS C:\Users\ASA> py c:\Users\ASA\OneDrive\Desktop\sample.py 10 52
2 of Diamonds
```